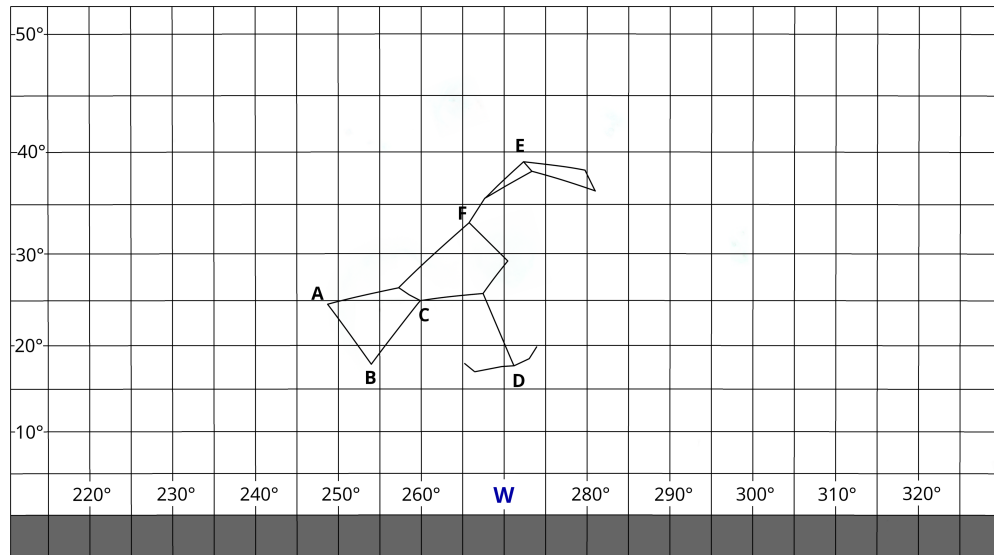


The given sky map (in Mercator projection) shows the constellation of Orion as seen from a certain location X (longitude $\lambda_X = 70^\circ$ E) on 21 March 2025 at 22:00 local time. The point “W” marks the cardinal West point. The altitude and azimuth values are marked on the grid.



(OM03.1) What is the approximate latitude (ϕ_X) of the location X?

(OM03.2) The grid provided on the Summary Answersheet has the same projection (Mercator), and